Investigating EEG Signals of Pediatric Patients with Intractable Seizures Using R

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Abstract:

Electroencephalography (EEG) provides a powerful noninvasive tool for monitoring brain activity, but rapid, nonlinear changes during seizures pose challenges for detection and prediction. This project explored EEG signals from the CHB-MIT Scalp EEG Database to distinguish seizure from non-seizure intervals and test machine learning approaches for forecasting signal patterns. Data preprocessing included scaling, smoothing with Kolmogorov–Zurbenko filters, and Butterworth bandpass filtering to reduce noise and highlight physiologically relevant frequencies. Statistical analyses of channel variance identified electrodes most affected during seizures, including FT10_T8, F8_T8, and F4_C4. Artificial neural networks (ANNs) were implemented to perform one-step-ahead prediction of EEG time series, with training on seizure and non-seizure segments. Results showed that ANN models converged more reliably on non-seizure intervals than on seizure episodes, reflecting the nonlinear instability of seizure dynamics. Smoothing improved model convergence compared to scaling alone, while runtime limitations highlighted the need for optimized model architectures and feature selection. These findings provide insights into EEG preprocessing and prediction challenges, establishing a foundation for future work in automated seizure detection. Future directions include integrating frequency-domain features, testing shorter overlapping windows, and extending ANN-based prediction toward real-time seizure forecasting.

Background/Objective:

- Seizures are defined as temporary deviations in the brain's electrical activity. They can result in very harmful effects, from brief losses of attention to full-body convulsions
- Regular seizure occurrences heighten risk of physical injury to patients and can potentially lead to death
- This database, collected at Children's Hospital Boston, contains EEG recordings from pediatric patients with intractable seizures
- After withdrawal of anti-seizure medication, patients' brain signals were monitored for several days to characterize seizures and evaluate their suitability for surgery
- The objective of this study is to take a closer look into patient 1, looking through each brain channel and predicting EEG signals.
- Patient 1 (file chb01) is an 11-year-old female. She experienced 7 seizures in: chb01_03.edf, chb01_04.edf, chb01_15.edf, chb01_16.edf, chb01_18.edf, chb01_21.edf, and chb01_26.edf (Each a different hour)

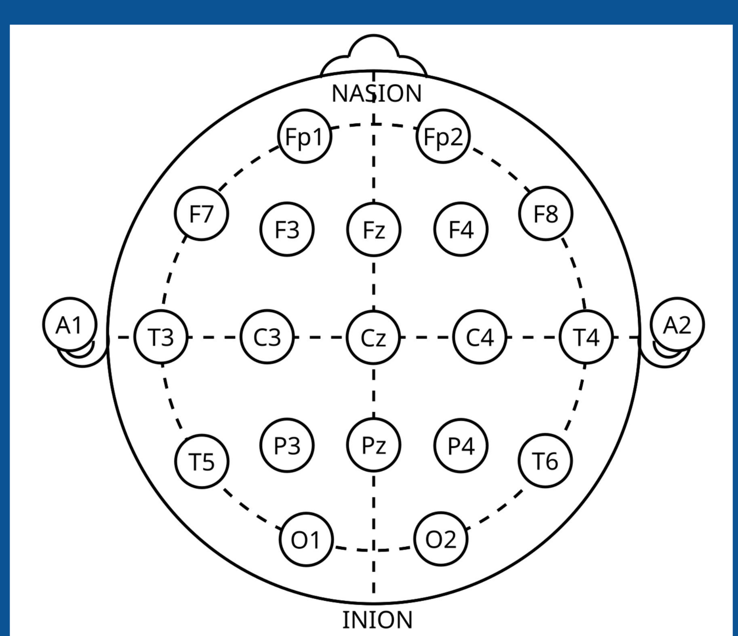

Figure 1: Diagram of the 23 electrodes on the brain. The difference between two electrode (a starting and ending point) signals measures the electricity flow between them. Ex: F7_T7 is the difference in voltage between F7 and T7

Published figure from: [21 Electrodes of International 10-20 System for EEG.](https://commons.wikimedia.org/wiki/File:21_electrodes_of_International_10-20_system_for_EEG.svg) [1]

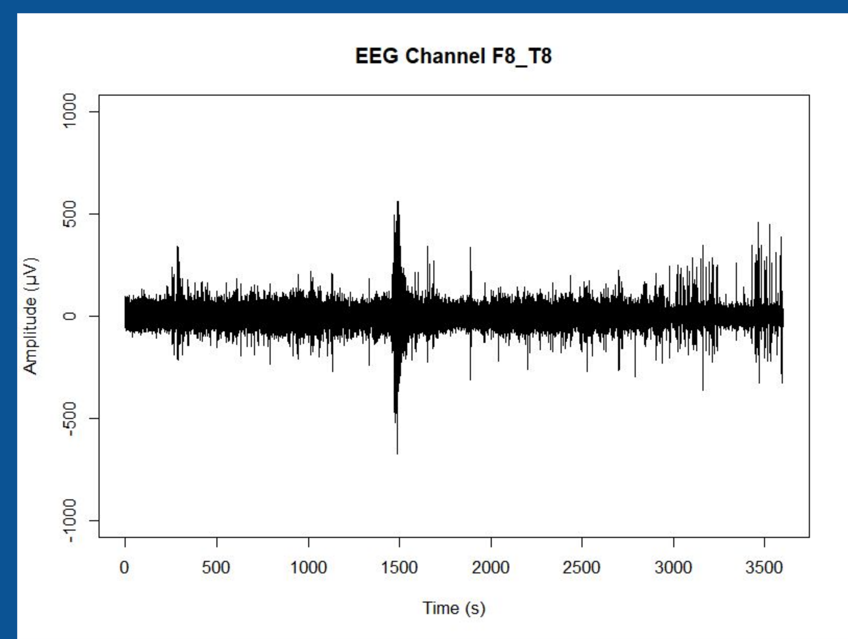

Figure 2: Graph of EEG Chanel F8_T8. The EEG shows 1 hour of raw data (chb01_04), with a seizure clearly visible at 1467–1494s, marked by large amplitude fluctuations.

- To compare seizure and non-seizure data, I calculated the standard deviation (by 5 second) of signal of chb01_01 (no seizure) and chb01_04 (with seizure) and plotted the T8_F8 channel
- The seizure at 1467–1494s in chb01_04 is clearly visible.

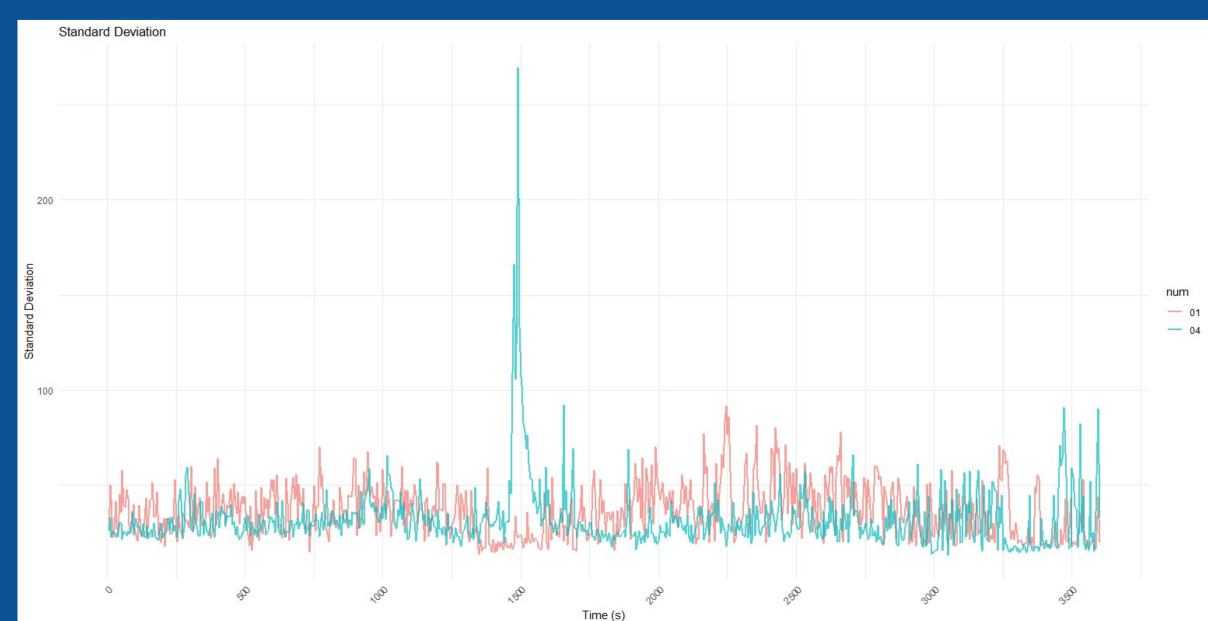

Figure 3: Graph of standard deviation of signal for the seizure (blue) vs. non seizure (red) data over time of T8_F8 channel.

Figure 4: Graph of standard deviation of signal for seizure (blue) vs. non seizure (red) data of patient 1 over time of all 23 channels. Most impacted channels include: FT10_T8, FT9_FT10, F8_T8, and F4_C4.

Figure 5: Graph of Patient 1's 7 seizure-affected data/hour (colored by hour) across all 23 channels, showing when seizures occur and which channels are most affected.

Data & Preprocessing:

Description + Goal: Training on the first 10 seconds of chb01_03 to create an ANN model and using the model to predict the next point after the first 10 seconds of chb01_04 (This is all done on non seizure data)

- Scale: I first scaled the data so that the voltage from each seizure/non seizure hour would be comparable.
- Smooth: Then, smoothing the data removes high-frequency noise that can destabilize training allowing for better algorithm convergence
- Window: First 10 seconds of chb01_03 and chb01_04 (non seizure)

Models & Parameters:

Originally, my model was unable to converge on 10 seconds of non seizure chb01_03 data after scaling and smoothing. To fix this I adjusted a few hyperparameters:

- Increasing stepmax to stepmax=1e+06 (higher complexity), the algorithm was able to converge, but slowed down run time by a lot (10s ran in around 5 mins)
- Increasing the threshold value to 0.1 decreased run time significantly

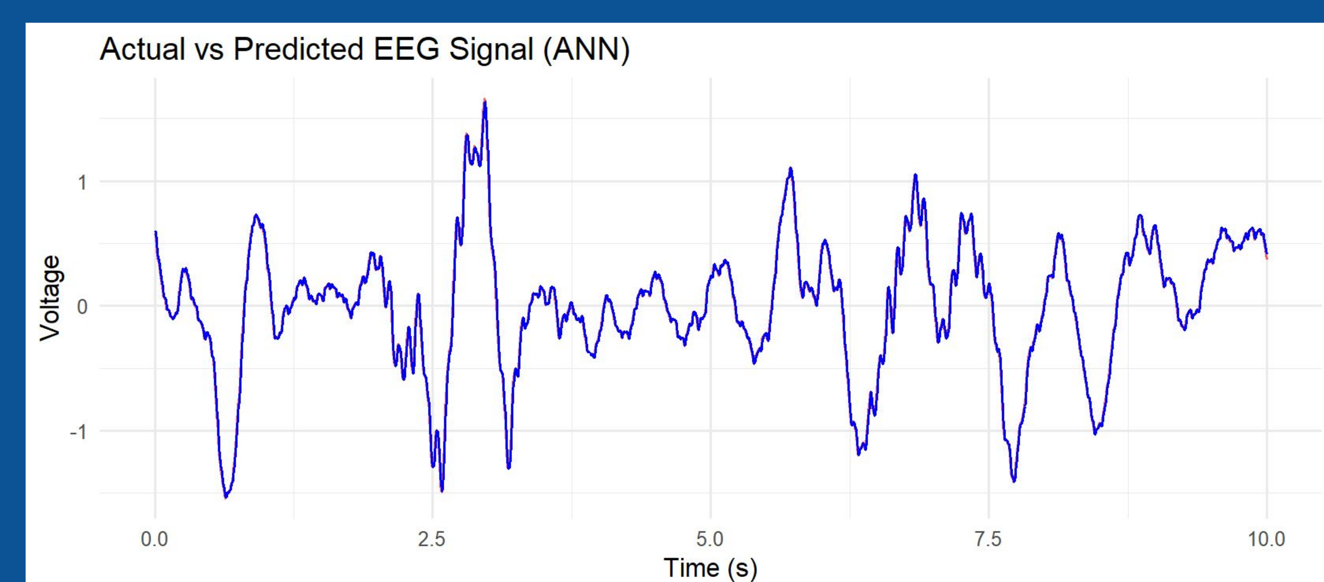

Figure 6: Graph of the actual data (blue) compared to the predicted (red) of the first 10 seconds of chb01_04 data using the prediction model trained on the first 10s of chb01_03 data

Discussion/Conclusion:

- Settings:
  - Scaling standardized the magnitude while smoothing removed high-frequency noise
  - Tuning hyperparameters of stepmax and threshold allowed for the model to converge on non seizure data and run a bit faster
- Limitations:
  - ANN model was unable to converge on seizure durations of the data since the sudden, nonlinear changes were harder for the model to capture
  - Run time was very slow. Training on 10s of non seizure data took up to 5 minutes
- Next step:
  - Predicting more data points at a time and being able to predict further into the future
  - Predicting on seizure intervals
  - Observing the differences in seizure pattern of the chb01 patient at 11-years-old and at 12.5-years-old
  - Using one patient channel to predict the data of a different patients

Takeaways:

- Visualizing seizure vs. non seizure intervals in an epileptic patient as well as most affected channels
- Improving ANN model stability by scaling data, smoothing data, and tuning hyperparameters
- Future directions include testing on seizure intervals, predicting further into the future, comparing patient signals across ages, and exploring cross-patient channel prediction.

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